

Object Oriented Programming

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Last Week

- Scope of variables
- Lifetime of variables
- Functions
 - Function Overloading

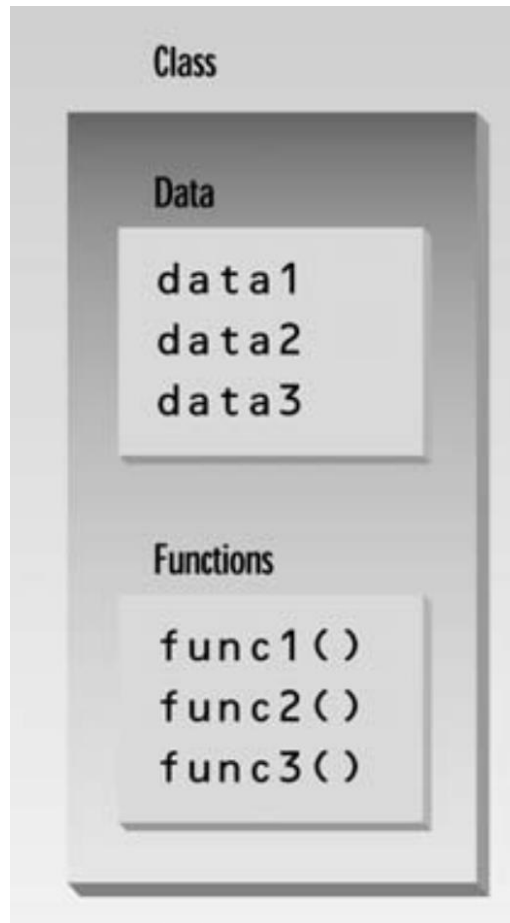
This Week

- Objects and classes

Classes

- Structures: way to group data elements
- Functions: way to organize program actions as named entities
- Classes: way to combine these ideas together
- **Placing data and functions together into a single entity is a central idea in OOP**

Class Diagram



class diagram in the [Unified Modeling Language](#) (UML) is a type of static structure diagram that describes the structure of a system by showing the system's [classes](#), their attributes, operations (or methods), and the relationships among objects.

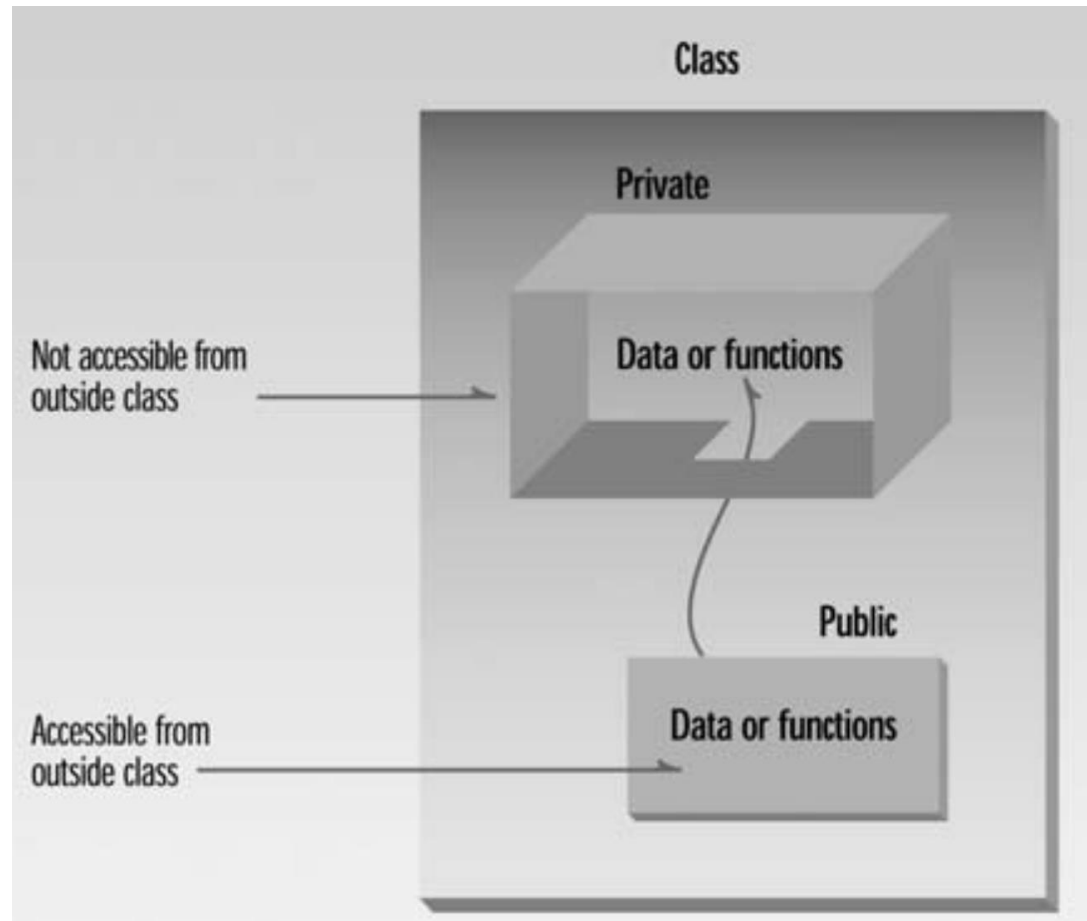
How does a class look like

```
1  class someClass
2  {
3      private:
4          int someData;
5
6      public:
7          void setData(int data)
8          {
9              someData = data;
10         }
11         void showData()
12         {
13             cout << "Data : " << someData << endl;
14         }
15     };
```

Terminology

- Private (data)
- Public (functions)
- data hiding?

Data hiding



Questions

- Can we make functions private?
- Can we make data public?

Practice

- Draw the class diagram of the last example.
- Update the class diagram when there is one more attribute, called “name” of type string in the class. Add member functions to set and show this attribute.
- Update the class code for this modified class diagram.

How to use this class?

```
17  int main()  
18  {  
19      SomeClass obj1;  
20      SomeClass obj2;  
21  
22      obj1.setData(3);  
23      obj2.setData(45);  
24  
25      obj1.showData();  
26      obj2.showData();  
27  }
```

Person class

- Draw the class diagram for a class Person. The data members of this class are name (string), age (int), weight (float). Functions members set and show these data attributes.
- Write the code for this class and also write the test program to test it.

Circle Class

- Draw class diagram of class Circle. Data member are xCord, yCord for center of circle, and radius. Member functions set, get and show this data. Also add a member function to calculate diameter, area and circumference
- Write c++ code for this class.

Special member functions

- Constructors
 - Automatically called when an object is created
- Destructors
 - Automatically called when an object is destroyed

Point class revisited

```
class Point
{
    int xCoord, yCoord;
public:
    Point() { xCoord = 0; yCoord = 0; }
    Point(int x, int y) { xCoord = x; yCoord = y; }

    ~Point() {}
};
```

Questions?

- Do you see any overloading in the example on last slide?

Distance Class

- Class diagram Distance which holds distance in feet and inches. Member functions set and clear data. Show the use of Constructor and destructor for this class. Add member function to get distance as input from user. Add member function to add feet (add_feet) and inches (add_inches). Add member function “add_distance” to add distance provided as an argument.

Summary

- Classes and objects are the primary concepts in OOP
- Private data and public members

Next Week

- Objects and classes